

lutions firm OpalForce—says India houses over 200 chip design and embedded software companies. However, it currently has a small pool of skilled workers.

Acknowledging this challenge, the All India Council for Technical Education (AICTE) launched a BTech Electronics VLSI Design & Technology and a Diploma in IC Manufacturing. VLSI stands for very large-scale integration design, and IC stands for integrated circuits. Many colleges have already rolled out the courses, including the premier Indian Institutes of Technology. MeitY aims to create a skilled workforce of more than 85,000 by 2027.

TECHNOLOGY TROUBLE

The applications filed with the Indian government to set up foundries must include a technology partner. Unlike in the past, where governments approved projects with deep financial backing, the current administration has made having manufacturing-grade technology (MGT) as a key qualifier. This has proven a big challenge.

There are very few MGT companies worldwide, like Intel, Samsung Electronics, Micron, STMicroelectronics, TSMC, Tower Semiconductor, and PSMC. The failure to get a technology partner was also the reason the Vedanta-Foxconn joint venture application filed in February 2022 was held up, and eventually that was the reason the JV fell through.

The technology transfer involved takes years. At the moment, no MGT major is interested in engaging in a multi-year saga of technology transfer to entities lacking adequate capabilities for enabling the transfer, adds Faruqui of Fab Economics.

“There are only six or seven companies that have MGT, and all of them have their hands full. They are

CONGLOMERATE FORAYS



Teamed up with Foxconn to apply for chip fab. JV split in July 2023, and Vedanta is scouting for tech partner



It intends to make a beginning with an ATMP unit and set up a foundry in the long run



Subsidiary Tarq Semiconductors has submitted revised proposals for ATMP facility and a compound semiconductor unit



Rumoured to have been interested in entering the chip race along with HCL via ISMC last year. ISMC proposal was put on hold



Plans to set up a subsidiary with an investment of ₹830 crore. It is looking to enter with fabless chip design



It is planning a \$791-million foray through CG Power and Industrial Solutions; has filed an application

willing to license but it's not simply about paper. It is about a group of 100 engineers that have to come [and] license technology to transfer it,” explains Chandrasekhar.

GREENFIELD SET-UP

After Covid-19 and during the acute phase of chip shortage, almost all large countries decided that semiconductor technology and manufacturing were the key strategic and economic assets. They launched schemes to attract semiconductor firms, like with the US's CHIPS Act. Since developed

countries already had an existing ecosystem, large projects with advanced technologies went there.

The Indian government is not unaware of this dynamic. “The US has been [making] semiconductors for decades, since I was a child. They have a mature semiconductor ecosystem, and the US CHIPS Act was simply about incentivising their expansion,” says Chandrasekhar.

But there are many other factors at play as well. Though the government is offering incentives, some industry experts say the infrastructure and operational costs for